

Design Documentation

16-bit Pipelined Wallace Tree Multiplier

Capstone Project Report

Department of Electronics and Communication Engineering

Jain (Deemed-to-be University), Bengaluru

1. Abstract

This document presents the design and implementation of a 16-bit Pipelined Wallace Tree Multiplier. The architecture combines a Wallace Tree reduction network with a 5-stage synchronous pipeline to achieve high-throughput unsigned 16-bit multiplication, producing a 32-bit product. The design is described in Verilog HDL, simulated using Icarus Verilog, and verified against 1012 test cases (including directed corner cases and 1000 random inputs), achieving a 100% pass rate with zero failures. The pipeline delivers one result per clock cycle at steady state, with a latency of 4 clock cycles.

2. Introduction

High-speed multipliers are fundamental building blocks in digital signal processing, microprocessors, and cryptographic hardware. The Wallace Tree algorithm reduces the number of partial product rows logarithmically using carry-save adders (full adders and half adders), significantly lowering the critical path compared to sequential or array multipliers. Pipelining the reduction network further improves clock frequency and system throughput by allowing a new set of operands to be accepted every clock cycle.

This design targets unsigned 16-bit operands and produces a 32-bit product. The architecture is structured as five pipeline stages separated by synchronous registers, with an active-low synchronous reset and a valid signal tracking mechanism to indicate when output data is meaningful.

3. Design Specifications

Parameter	Value
Architecture	Wallace Tree with 6 reduction levels
Operand Width	16 bits (unsigned)
Product Width	32 bits
Pipeline Stages	5 (4-cycle latency)
Throughput	1 result per clock cycle (steady state)
Clock Frequency	50 MHz (20 ns period)
Reset Type	Active-low synchronous (rst_n)
HDL	Verilog HDL
Simulation Tool	Icarus Verilog (iverilog)

Waveform Viewer	GTKWave
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4. Top-Level Port Description

Module: pipelined_wallace_mult_16x16

Signal	Width	Direction	Description
clk	1	Input	System clock (50 MHz)
rst_n	1	Input	Active-low synchronous reset. Clears all pipeline registers on the next rising clock edge.
valid_in	1	Input	Assert high when operands A and B are valid. Propagates through the valid pipeline alongside data.
A[15:0]	16	Input	Multiplicand (16-bit unsigned)
B[15:0]	16	Input	Multiplier (16-bit unsigned)
P[31:0]	32	Output (reg)	Final 32-bit product, registered. Valid when valid_out is asserted.
valid_out	1	Output (reg)	Asserted high when P holds a valid multiplication result.

5. Architecture Overview

The multiplier is organized as a five-stage pipeline. Each stage is a purely combinational block followed by a bank of synchronous registers. A new pair of operands may be applied on every rising clock edge. Once the pipeline is filled (after 4 cycles of latency), a valid product is produced on every subsequent clock cycle.

5.1 Pipeline Stage Summary

Stage	Module	Function
1	partial_product_gen	Generates 16 partial product rows (pp_row0 to pp_row15) using AND gates. Row $i = A \& \{16\{B[i]\}\}$.
2	wallace_stage1	Wallace Tree reduction levels 1 and 2. Reduces 16 rows using full adders and half adders.
3	wallace_stage2	Wallace Tree reduction levels 3 and 4. Continues compressing the carry-save columns.
4	wallace_stage3	Wallace Tree reduction levels 5 and 6. Every column reduced to at most 2 bits.
5	cla_32bit	32-bit Carry Lookahead Adder. Final addition of two remaining rows to produce P[31:0].

5.2 Sub-module Descriptions

partial_product_gen: A purely combinational module. Implements $pp_row_i = A \& \{16\{B[i]\}\}$ for $i = 0$ to 15, producing 16 independent 16-bit partial product rows. No adder logic is present; row shifting is handled implicitly by column indexing in the Wallace stages.

wallace_stage1 / wallace_stage2 / wallace_stage3: Each stage takes carry-save column vectors from the previous stage via pipeline registers and applies full adders and half adders to reduce column heights. A full adder maps three inputs to a sum and carry, reducing 3 bits to 2. After stage 4 (wallace_stage3), every column contains at most 2 bits.

cla_32bit: An 8-block, 32-bit Carry Lookahead Adder. Each 4-bit block computes block-level generate (BG) and propagate (BP) signals, allowing all inter-block carries to be resolved in parallel rather than rippling serially.

full_adder / half_adder: Primitive combinational cells used by the Wallace reduction stages. The full adder computes $Sum = A \oplus B \oplus Cin$ and $Cout = \text{majority}(A, B, Cin)$. The half adder computes $Sum = A \oplus B$ and $Cout = A \text{ AND } B$.

6. Adder Cell Count

The table below provides a complete count of full adder (FA) and half adder (HA) instantiations per Wallace reduction stage, as extracted directly from the Verilog source files.

Module	Full Adders	Half Adders	Total Cells	Wallace Levels
wallace_stage1	124	21	145	1 and 2
wallace_stage2	53	23	76	3 and 4
wallace_stage3	19	35	54	5 and 6
Total	196	79	275	6 levels

Note: The higher half adder count in wallace_stage3 compared to full adders is expected, as column heights have already been reduced to 2 or 3 bits by that stage, making half adders sufficient for most columns.

7. Pipeline Register Count

The table below enumerates the flip-flop resources used across all five pipeline stage boundaries. All registers are contained within the top-level module pipelined_wallace_mult_16x16 and are updated synchronously on the positive clock edge.

Register Bank	Signal Group	Flip-Flop Count	Notes
Stage 1 -> Stage 2	pp_row0_q to pp_row15_q	256	16 registers x 16 bits each
Stage 2 -> Stage 3	s2_colX_bY_q	116	1-bit registers, Wallace L1+L2 outputs
Stage 3 -> Stage 4	s3_colX_bY_q	73	1-bit registers, Wallace L3+L4 outputs
Stage 4 -> Stage 5	s4_colX_bY_q	68	1-bit registers, Wallace L5+L6 outputs

Output Register	P[31:0]	32	Final 32-bit product register
Valid Pipeline	valid_stage1..4, valid_out	5	1-bit shift register for valid tracking
Total		550	Flip-flops across all pipeline stages

Note: The register count decreases from Stage 1->2 to Stage 4->5 as the Wallace Tree progressively reduces the number of carry-save bits per column. The 256 flip-flops at Stage 1->2 reflect the full 16x16-bit partial product array; by Stage 4->5 only 68 single-bit registers are needed as every column is reduced to at most 2 bits.

8. Verification

8.1 Testbench Structure

Module: tb_pipelined_wallace_mult_16x16. The testbench is fully self-checking and pipeline-latency-aware. It uses a scoreboard (queue) to store expected products for each input pair applied to the DUT, matched against the output when valid_out is asserted 4 cycles later. The simulation was run using Icarus Verilog (iverilog) and waveforms were inspected in GTKWave.

8.2 Test Strategy

- Directed corner cases: 0x0, FFFFhex x FFFFhex, 0001hex x FFFFhex, 8000hex x 8000hex, and other boundary values.
- 1000 random test vectors: pseudo-random 16-bit A and B pairs generated using Verilog .
- Latency verification: the scoreboard ensures each output is compared against its correct corresponding input, not the simultaneous input.
- Waveform dump: a .vcd file is generated for post-simulation inspection in GTKWave.

8.3 Simulation Results

Metric	Result
Total results checked	1012
Passed	1012
Failed	0
Overall verdict	PASS
Pipeline latency (verified)	4 clock cycles
Simulation duration	~20.61 ms simulated time

Sample verified transactions from simulation output:

- A = 0xD609, B = 0x5663, Expected P = 0xFFFFC0004, Got P = 0xFFFFC0004 — PASS
- A = 57325, B = 34029, Expected P = 1950712425, Got P = 1950712425 — PASS
- A = 0xFFFF, B = 0xFFFF, Expected P = 0xFFFFE0001, Got P = 0xFFFFE0001 — PASS

9. Design File Manifest

File	Description
pipelined_wallace_mult_16x16.v	Top-level module; all pipeline registers and sub-module instantiations.
partial_product_gen.v	Stage 1: 16x16 AND partial product generation.
wallace_stage1.v	Stage 2: Wallace reduction levels 1 and 2 (124 FA, 21 HA).
wallace_stage2.v	Stage 3: Wallace reduction levels 3 and 4 (53 FA, 23 HA).
wallace_stage3.v	Stage 4: Wallace reduction levels 5 and 6 (19 FA, 35 HA).
cla_32bit.v	Stage 5: 32-bit Carry Lookahead Adder (8 x 4-bit blocks).
full_adder.v	Primitive full adder cell.
half_adder.v	Primitive half adder cell.
tb_pipelined_wallace_mult_16x16.v	Self-checking testbench with scoreboard-based latency-aware verification.
sim_output.txt	Simulation log with per-transaction PASS/FAIL and final summary.
wallace_mult_waveform.vcd	VCD waveform file for GTKWave post-simulation inspection.

10. Conclusion

The 16-bit Pipelined Wallace Tree Multiplier has been successfully designed, implemented in Verilog HDL, and functionally verified using Icarus Verilog with waveform analysis in GTKWave. The design employs 275 adder cells (196 full adders and 79 half adders) across six Wallace reduction levels, and 550 flip-flops distributed across five pipeline stage boundaries. It achieves a 4-cycle pipeline latency, single-cycle throughput at steady state, and correct operation across all 1012 test vectors including boundary conditions and 1000 random cases. The modular hierarchy, explicit pipeline register naming, and synchronous active-low reset make the design readily synthesisable on FPGA targets.